# SQL Fundamentals

## Introduction

Before learning SQL commands, it is important to understand some basic concepts:

- Data
- Database
- DBMS
- RDBMS
- SQL
- Difference between DBMS and RDBMS

Let us understand each concept using simple examples.

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# 1. What is Data?

## Definition

Data is a collection of raw facts, figures, or information that can be stored and processed.

## In Simple Terms

Data means individual pieces of information.

For example:

- Name: Sravya
- Age: 25
- Course: Data Science
- Marks: 85

Here, Sravya, 25, Data Science, and 85 are individual pieces of data.

## Real-Life Examples of Data

- Student Name
- Employee ID
- Phone Number
- Product Price
- Salary
- Marks
- Address

## Example

Suppose we have the following student information:

| Student ID | Name | Course | Marks |
| --- | --- | --- | --- |
| 101 | Sravya | Data Science | 85 |

Each individual value in the above table is called data.

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# 2. What is a Database?

## Definition

A database is an organized collection of data that can be easily stored, accessed, managed, and updated.

## In Simple Terms

Imagine a college has 1,000 students.

The college needs to store information such as:

- Student ID
- Student Name
- Age
- Course
- Marks

Instead of storing this information randomly, the college organizes all the information properly.

This organized collection of data is called a database.

## Example

| Student ID | Name | Course | Marks |
| --- | --- | --- | --- |
| 101 | Sravya | Data Science | 85 |
| 102 | Ravi | Python | 90 |
| 103 | Priya | SQL | 88 |

This organized collection of student information can be called a Student Database.

## Real-Life Examples of Databases

### Bank Database

A bank database may contain:

- Customer Details
- Account Details
- Transaction Details
- Loan Details

### Hospital Database

A hospital database may contain:

- Patient Details
- Doctor Details
- Appointment Details
- Treatment Details

### Shopping Database

An online shopping database may contain:

- Customer Details
- Product Details
- Order Details
- Payment Details

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# 3. What is DBMS?

## Full Form

DBMS stands for Database Management System.

## Definition

DBMS is software that helps users create, store, retrieve, update, and delete data from a database.

## In Simple Terms

Suppose a college has a database containing information about 1,000 students.

The college may want to:

- Add a new student
- View student details
- Update student marks

- Delete student details

Software is required to perform these operations.

That software is called a Database Management System or DBMS.

## Real-Life Example

Think about the Gallery application on a mobile phone.

The photos are the data.

The collection of photos is similar to a database.

The Gallery application allows us to:

- Add photos
- View photos
- Edit photos
- Search for photos
- Delete photos

Similarly, a DBMS helps us manage the data stored inside a database.

## CRUD Operations

The four basic operations performed on data are called CRUD operations.

CRUD stands for:

- C - Create
- R - Read
- U - Update
- D - Delete

### Example

- Create - Add a new student
- Read - View student information
- Update - Change student marks
- Delete - Remove a student record

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# 4. What is RDBMS?

## Full Form

RDBMS stands for Relational Database Management System.

## Definition

RDBMS is a type of Database Management System that stores data in the form of tables containing rows and columns and allows relationships between tables.

## In Simple Terms

In an RDBMS, data is stored in tables.

Multiple tables can be connected to each other using common columns.

For example, consider two tables.

## Students Table

| Student ID | Student Name | Course ID |
|---|---|---|
| 101 | Sravya | C01 |
| 102 | Ravi | C02 |

## Courses Table

| Course ID | Course Name |
|---|---|
| C01 | Data Science |
| C02 | Python |

The Students table and Courses table both contain Course ID.

Therefore, the two tables can be connected using Course ID.

This relationship between tables is an important feature of RDBMS.

## Examples of RDBMS

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

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# 5. What is SQL?

## Full Form

SQL stands for Structured Query Language.

## Definition

SQL is a language used to communicate with relational databases and perform operations on data.

## In Simple Terms

SQL is the language we use to give instructions to a database.

For example, we can use SQL to:

- Create a database
- Create a table
- Add data
- View data
- Update data
- Delete data

## Example 1: Viewing Data

```sql
SELECT * FROM students;
```

The above SQL command tells the database:

"Show me all the information from the students table."

## Example 2: Adding Data

```sql
INSERT INTO students
VALUES (101, 'Sravya', 'Data Science', 85);
```

The above command tells the database:

"Add this student information to the students table."

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# 6. Difference Between DBMS and RDBMS

| DBMS | RDBMS |
|---|---|
| DBMS stands for Database Management System | RDBMS stands for Relational Database Management System |
| DBMS is used to store and manage data | RDBMS stores and manages data in tables |
| Data may not have relationships | Tables can have relationships with each other |
| Relationships are not the main concept | Relationships between tables are an important concept |

| DBMS | RDBMS |
|---|---|
| Generally suitable for smaller amounts of data | Suitable for handling large amounts of structured data |
| May have limited support for keys | Uses Primary Keys and Foreign Keys |
| Provides basic data management | Provides better data consistency using relationships and constraints |

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# 7. Connection Between Data, Database, DBMS, RDBMS, and SQL

The easiest way to understand all the concepts is:

Data

↓

Database

↓

DBMS

↓

RDBMS

↓

SQL

## Explanation

### Data

Individual information such as:

- Sravya
- 25
- Data Science
- 85

↓

**Database**

An organized collection of student information.

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**DBMS**

Software used to manage the database.

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**RDBMS**

A type of DBMS where data is stored in tables and tables can be related to each other.

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**SQL**

The language used to communicate with the RDBMS.

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# 8. Simple Real-Life Example

Consider a college.

## Data

Student name, age, marks, and course are data.

## Database

The collection of all student records is a database.

## DBMS

The software used to manage student information is a DBMS.

## RDBMS

Student, Course, and Teacher information is stored in different tables, and the tables are connected to each other.

This is an RDBMS.

### SQL

Commands used to add, view, update, and delete student information are written using SQL.

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## 9. Quick Revision

| Concept | Simple Meaning |
|---|---|
| Data | Individual facts or information |
| Database | Organized collection of data |
| DBMS | Software used to manage databases |
| RDBMS | DBMS that stores data in related tables |
| SQL | Language used to communicate with relational databases |

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## 10. Key Points to Remember

- Data means raw facts and information.
- A database is an organized collection of data.
- DBMS is software used to manage databases.
- CRUD stands for Create, Read, Update, and Delete.
- RDBMS stores data in tables.
- Tables in an RDBMS can be related to each other.
- SQL is used to communicate with relational databases.
- MySQL, PostgreSQL, Oracle Database, and Microsoft SQL Server are examples of RDBMS.

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## Conclusion

Understanding Data, Database, DBMS, RDBMS, and SQL is the first step in learning SQL.

The basic learning flow is:

Data → Database → DBMS → RDBMS → SQL

After understanding these fundamentals, the next step is to learn about databases, tables, rows, columns, records, fields, and SQL data types.